

Customer Retention & Churn Analysis Report

Data Science & Analytics Internship Submission – Future Interns

Executive Summary

This report analyzes customer churn behavior in a subscription-based telecom business. The objective is to identify key churn drivers, evaluate retention trends, estimate Customer Lifetime Value (CLV), and provide actionable recommendations to reduce customer attrition and improve long-term profitability.

Key KPIs

- Churn Rate (%) & Retention Rate (%)
- Average Monthly Charges
- Average Customer Tenure
- Estimated Customer Lifetime Value (CLV)
- Churn by Contract Type
- Churn by Tenure Group

Key Findings

• Month-to-month contract customers show significantly higher churn compared to long-term contracts. • Customers within their first 12 months have the highest churn probability. • Higher monthly charges are associated with increased churn risk. • Customers with longer tenure demonstrate stronger loyalty and higher lifetime value.

Retention & CLV Insights

Tenure-based analysis reveals that churn peaks during the first year and stabilizes after 24 months. Customer Lifetime Value (CLV) was estimated using: $CLV = \text{Average Monthly Charges} \times \text{Average Tenure}$. Retained customers contribute significantly higher lifetime revenue than churned customers, highlighting the financial importance of improving early-stage retention.

Strategic Recommendations

1. Encourage long-term contracts through loyalty incentives and bundled offers. 2. Improve onboarding experience for customers in the first 12 months. 3. Provide retention-focused discounts for high monthly charge customers. 4. Implement churn prediction monitoring for proactive intervention.

Conclusion

This analysis identifies key churn drivers and revenue opportunities. By optimizing contract strategy, pricing, and early customer engagement, the company can significantly reduce churn and enhance long-term profitability. This project demonstrates applied retention analytics, cohort analysis, lifetime value estimation, and strategic business decision-making.